Unit: Unit 0: Foundations of Algebra
Topic: Inequalities

Name: ____________
Date: ________

Challenge (Habanero)

Q1. A hiker can travel x miles in 3 hours. If she travels for 5 hours, how many miles d can she cover if $d > 2x - 5$?
(A) $d > 5$
(B) $d > 10$
(C) $d > 15$
(D) $d > 20$
(E) $d > 25$

Q2. Solve the inequality $2x - 5 > 11$ and graph the solution on a number line.
(A) $x > 8$
(B) $x < 8$
(C) $x > 3$
(D) $x < 3$
(E) $x = 8$

Q3. A fuel tank can hold 15 gallons of gas. If 3 gallons are already in the tank, how many more gallons g can be added if $g < 2(15 - 3)$?
(A) $g < 12$
(B) $g < 15$
(C) $g < 18$
(D) $g < 20$
(E) $g < 24$

Q4. The population of rabbits in a forest is given by $P = 2x+5$, where x is the number of months since the initial count. If $P > 25$, what is the range of x?
(A) $x > 5$
(B) $x < 5$
(C) $x > 10$
(D) $x < 10$
(E) $x = 5$

Q5. Write the solution to the inequality $x - 2 > 7$ in interval notation.
(A) $(9, \infty)$
(B) $(-\infty, 9)$
(C) $(7, \infty)$
(D) $(-\infty, 7)$
(E) $[9, \infty)$

Q6. A park ranger can travel 20 miles in 4 hours. If she travels for 6 hours, how many miles d can she cover if $d > 3(20-4x)$ and $x = 2$?
(A) $d > 30$
(B) $d > 40$
(C) $d > 50$
(D) $d > 60$
(E) $d > 70$

Q7. Solve the inequality $\frac{2x}{3} + 2 > 8$.
(A) $x > 9$
(B) $x < 9$
(C) $x > 12$
(D) $x < 12$
(E) $x = 9$

Q8. A group of friends want to hike a trail that is 15 miles long. If they have already hiked x miles, how many more miles m can they hike if $m < 2(15 - x)$?
(A) $m < 10$
(B) $m < 15$
(C) $m < 20$
(D) $m < 25$
(E) $m < 30$

Open Ended Questions (FRQ)

Q9. The population of a certain animal species is given by $P = 5x + 2$, where x is the number of years since the initial count. If $P > 50$, find the range of x and graph the solution on a number line.

Q10. A car's fuel efficiency is given by 25 miles per gallon. If the car travels x miles, how many gallons of gas g are needed if $g > \frac{x}{25}$? Write the solution in interval notation.

Chef's Tips

- Tip 1: Emphasize the importance of checking the endpoints of the solution intervals.
- Tip 2: Have students use number lines to visualize the solutions to the inequalities.
- Tip 3: Encourage students to use algebraic methods to solve the inequalities, rather than just graphical methods.